GRIFFITH UNIVERSITY

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

3806ICT Logic and Automated Reasoning

Trimester 1, 2026

Assignment 1 (30%)

Due Time: 23:59 Friday, 1st May 2026

Assignment overview and requirements

This is an individual assignment.

Your goal is to implement *and improve* the automated reasoning algorithm Algorithm 2 on Page 67 of the textbook (covered in lectures around Weeks 4 - 5), and test your implementation on a set of benchmark datasets.

More specifically, you are expected to:

- Write a parser that reads first-order logic formulae text (from file) as input and outputs the formulae in a format supported by your downstream pipeline.
  - The input format could be a text file that contains one formula per line in the syntax used in this course.
  - Alternatively, you can also use industry standards, such as SMT-LIB and TPTP, as input formats.
  - The output format could be anything that your program can understand and use.
- Implement Algorithm 2 on Page 67 of the textbook as the "baseline" method.
- Brainstorm ways to improve the performance (in terms of how many formulae it can automatically solve and how long it takes) of Algorithm 2. Document and implement your improvements.
  - Your improved algorithm cannot be a wrapper of an existing solver. It must be an actual improvement of the baseline algorithm.
- Find a set of suitable benchmark datasets online or generate some datasets.
  - If you use datasets online, cite the source.
  - If you generate datasets, document the generation process and justify that the dataset is "suitable":
    - It should contain a balanced set of easy and hard formulae that can put your method to a serious test.
- Conduct comprehensive evaluation of your method (compare it with the baseline) using the datasets and document your findings.

Everything not specified above are up to you. These include:

- the programming language,
- the libraries and existing code you may use,
- how you improve the baseline,
- how you use AI to help you, except for writing.

Instructions for writing the report

- The report should be in LNCS format. Do not change font size, margin size etc.
    - Word template can be found here:

https://www.springer.com/gp/computer-science/lncs/conference-proceedings-guidelines

    - LaTex template can be found on Overleaf:

https://www.overleaf.com/latex/templates/springer-lecture-notes-in-computer-science/kzwwpvhwnvfj

    - The report should be *no more than* **8 pages, including references**.
- Your report should have an abstract.
    - No more than 300 words.
    - Give an overview of the topic and what you have learned.
- Your report should have an introduction section.
    - Describe the background of the topic and why it's important.
    - Give an overview of the state-of-the-art/related methods.
    - Give an overview of your work.
- Your report should have a "proposed approach" section.
    - Describe your work in detail.
    - Focus on the conceptual design.
- Your report should have an implementation and experiment section.
    - Describe the implementation details (e.g., programming language, libraries, important programming specifics, but no need to show code unless it is significant.)
    - Describe the experiment setup (e.g., OS, hardware specs, etc.).
    - Discuss and analyse the results.
- Your report should have a discussion/conclusion section at the end.
    - Summarise the most significant ideas you have learned.
- Your report should have a "data availability" section of just one sentence that provides the link to your source code.
- The order of sections is up to you.
- Use figures and tables to present your ideas.
    - Vectorised images are preferred. If other images (e.g., screenshots) are used, ensure that they do not look blurry/jagged on a 200+ PPI (e.g., 4K 21-inch or 5K 27-inch) screen.
- Cite existing work, including all the libraries and code you used that are not developed by yourselves.
    - The reference should contain the authors, the paper/project title, the year of publication, and the venue of publication (name of conference/journal).

    - In the case of code or library, venue of publication is optional, but provide URL.
- Acknowledge the usage of generative AI. Document the queries and responses in an Appendix (exclude from page limit). Do not use generative AI for writing.

SUBMISSION

Each student submits ONE PDF copy of the report to L@G (Canvas) / 3806ICT / Assessments / Assignment 1

Late submissions: marks will be deducted according to university policy.

References

[1] Zhe Hou. Fundamentals of Logic and Computation: With Practical Automated Reasoning and Verification. Springer (Texts in Computer Science), 2021.